The Kingman Coalescent: An All–n Genealogy and Branch-Length Atlas

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Abstract

We treat the partition-valued Kingman coalescent as one mathematical owner. Unit rate per unordered block pair gives a pure-death block count with $\lambda_k = \binom{k}{2}$. Partial fractions close every hypoexponential transition, independent holding times close the MRCA law, and a projective coupling gives finite infinite-sample absorption. Scaling by the number of branches identifies total tree length with the maximum of iid Exp(1/2) variables, yielding an exact CDF. Markov determinants are not Artin–Mazur zeta functions.

1 Partition owner and block chain

For each $n \geq 1$, start from the discrete partition of $[n]$ and merge each unordered pair of current blocks at rate one. The labelled partition process is the Kingman coalescent. Its block count K_t starts at n and has

$$\lambda_k = \binom{k}{2} = k(k-1)/2, \qquad Q_{k,k-1} = \lambda_k, \quad Q_{k,k} = -\lambda_k,$$

with state 1 absorbing. Conditional on k blocks, the next pair is uniform. The family is coupled by restricting a partition on $[n+1]$ to $[n]$.

For $1 \leq j \leq i$ the all–n transition law is hypoexponential:

$$p_{ij}(t) = \left(\prod_{m=j+1}^{i} \lambda_m\right) \sum_{\ell=j}^{i} \frac{e^{-\lambda_\ell t}}{\prod_{m=j,m\neq\ell}^{i}(\lambda_m - \lambda_\ell)}. \tag{1}$$

The empty-product convention gives $p_{ii}(t) = e^{-\lambda_i t}$ and the $i = 1$ absorbing row.

The formula can also be read as a semigroup certificate. Its Laplace transform is

$$\int_0^\infty e^{-st} p_{ij}(t)\, dt = \frac{\prod_{m=j+1}^{i} \lambda_m}{\prod_{m=j}^{i}(s + \lambda_m)},$$

and direct multiplication of these triangular kernels gives $P(t+u) = P(t)P(u)$. Thus no independently fitted transition table is needed.

2 MRCA time and projective limit

Let $E_k \sim \mathrm{Exp}(\lambda_k)$ be the holding time at level k. The pair-clock construction and memorylessness make the E_k independent of one another and of the uniform merger choices. Thus

$$T_n = \sum_{k=2}^{n} E_k, \qquad M_n(s) := \mathbb{E}[e^{-sT_n}] = \prod_{k=2}^{n} \frac{\lambda_k}{\lambda_k + s}.$$

Consequently,

$$\mathbb{E}[T_n] = \sum_{k=2}^{n} \frac{1}{\lambda_k} = 2\left(1 - \frac{1}{n}\right), \qquad \mathrm{Var}(T_n) = \sum_{k=2}^{n} \frac{1}{\lambda_k^2}. \tag{2}$$

The restriction coupling is essential for the infinite statement. It gives $T_n \le T_{n+1}$ on one probability space, hence $T_n \uparrow T_\infty$ almost surely. Summability of the means and variances implies finite absorption and

$$\mathbb{E}[T_\infty] = 2, \qquad \mathrm{Var}(T_\infty) = 4\big(2\zeta(2) - 3\big) = \frac{4\pi^2}{3} - 12,$$

with Laplace transform the convergent product obtained by sending $n \to \infty$. This is a projective-coupling assertion, not a claim that independent marginal sums share a sample path.

3 Total branch length

Define $L_n = \int_0^{T_n} K_t\,dt = \sum_{k=2}^n kE_k$. Since scaling an exponential by k changes its rate to $(k-1)/2$,

$$\mathbb{E}[e^{-sL_n}] = \prod_{j=1}^{n-1} \frac{j/2}{j/2 + s}, \qquad \mathbb{E}[L_n] = 2H_{n-1}, \qquad \mathrm{Var}(L_n) = 4H_{n-1}^{(2)}.$$

Let $Y_1, \ldots, Y_{n-1}$ be iid Exp(1/2) variables. Their order-statistic spacings, read from the largest backwards, have rates $(n-1)/2, (n-2)/2, \ldots, 1/2$. Therefore the sum above has the same law as $\max_i Y_i$, and the exact CDF is

$$\Pr\{L_n \le \ell\} = \big(1 - e^{-\ell/2}\big)^{n-1}, \qquad \ell \ge 0. \tag{3}$$

At $n = 1$, $L_1 = T_1 = 0$ and the convention in (3) is the constant one.

For orientation, the first Bell numbers are

$$1,\ 2,\ 5,\ 15,\ 52,\ 203,\ 877,\ 4140$$

for $n = 1, \ldots, 8$; they are used only as an independent finite partition ledger, not as an approximation to the infinite process.

4 Certificate and source boundary

The evidence payload contains 312 transition rows through $n = 12$, 12 holding rows, 48 MRCA rows, 60 branch rows, and an independent Bell-number partition ledger through $n = 8$. The checker recomputes partial fractions, row sums, Chapman–Kolmogorov, moments, the maximum CDF, and the $n = 1$ boundary. SymPy independently verifies the rational transforms and beta-integral behind (3); replay and hostile mutations test reproducibility and schema closure.

The construction is attributed to Kingman (1982), without a priority or novelty claim. It has no intrinsic rational-prime carrier, primitive orbit clock, or arithmetic divisor. The strict Route-A verdict is

`(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FAIL)`,   `ROUTE_A_REJECTED`.

In particular, a finite Markov determinant, trace-log, or Laplace product is not an Artin–Mazur dynamical zeta and is not used as one.

References

[1] J. F. C. Kingman, "The coalescent," *Stochastic Processes and their Applications* 13, 235–248 (1982), doi:10.1016/0304-4149(82)90011-4.